

EXECUTIVE STRATEGIC DIFFERENTIATION BRIEFING · 30-SECOND JURY OVERVIEW

WHY THIS IS DIFFERENT

"From environmental mapping to explainable spatial risk intelligence"

Core Positioning: An evidence-oriented GeoAI prototype that turns heterogeneous environmental observations into an interpretable spatial risk-screening workflow — while explicitly separating demonstrated evidence from research reconstruction and future deployment.

01 PRIORITIZES, NOT JUST DISPLAYS

"From visual monitoring to structured spatial prioritization."

Conventional deforestation dashboards stop at visualizing where forest loss has already occurred. This platform organizes multi-source spatial signals into an actionable relative risk-scoring and screening workflow across ~150,000 precomputed spatial grid cells in Rondônia.

~150,000 Precomputed Cells

Relative Spatial Risk Score

Screening Filter

02 FUSES SPATIAL EVIDENCE

"Reconstructing multi-source drivers instead of mapping a single layer."

Rather than treating a single satellite layer as the complete answer, the research pipeline reconstructs environmental predictors across 4 physical groups: infrastructure accessibility, historical frontier momentum, demographic pressure, and topographic terrain constraints.

Road Proximity

Historical Loss Distance & Density

Population Pressure

Elevation

03 SEPARATES TWO EVIDENCE TRACKS

"Documented production benchmark vs. independent research evaluation."

Maintains strict scientific separation between the documented original production benchmark and the independent Model V2 research evaluation. They are distinct empirical experiments on different populations; conflating them is formally rejected as unscientific.

Track 1: ROC-AUC 0.82 (Ref)

Track 2: ROC-AUC 0.7474 (2024 Test)

5.0x Retrospective Top-Decile Lift

04 EXPLAINS THE RISK SIGNAL

"Not only where risk is higher, but which spatial signals explain the ranking."

Combines 25 years of official INPE PRODES history (2001–2025 across all 9 Legal Amazon states), exact TreeSHAP marginal feature attributions for Model V2 (41.3% historical loss proximity), documented regional macro-drivers (41.0% roads), and interactive sensitivity sliders.

TreeSHAP Attribution (Model V2)

Documented Macro Importance

25-Yr PRODES History

05 BUILT WITH SCIENTIFIC BOUNDARIES

Evidence Integrity & Discipline

"Documented scientific boundaries framed as a strength, not an apology."

The platform explicitly distinguishes demonstrated prototype evidence, reconstructed research, assumptions, and future operational prerequisites. When original training seeds and binary labels were unrecorded upstream, **benchmark reproduction was intentionally halted** rather than manufacturing deceitful pseudo-labels from model outputs. Outputs are relative spatial rankings, not calibrated probabilities.

Scientific Halt Notice (Zero Fake Labels)

Relative Rankings (Not Calibrated)

Offline XGBoost (No Live Telemetry Claims)

99 Deterministic Tests (100% Pass)

DISTINCTIVE ARCHITECTURE: CONVENTIONAL MONITORING VS. GEOAI INTELLIGENCE

SYSTEM PARADIGM SHIFT

CONVENTIONAL DASHBOARD

Monitor → Visualize
Passive historical loss display

THIS PROJECT (GEOAI DECISION SUPPORT)

Observe → Engineer Features → Rank Risk → Explain Signals → Explore Scenarios → Plan Verification
Multi-source feature engine · Relative spatial screening · TreeSHAP explainability · Decision support

"The differentiation is not a claim of replacing established forest-monitoring systems; it is the integration of spatial risk ranking, historical context, explainability, and evidence discipline into one decision-support prototype."